Multimodal Emotion Recognition using Speech, Text, and Speech-Text Fusion

Project Report

Submitted by

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GitHub Repository

`https://github.com/savioshaju/Multimodal-Emotion-Recognition.git`

Project Outputs and Model Checkpoints

`https://drive.google.com/drive/folders/`
`1Y0OlP0rBU5A2lPE7DFwtUskw0kB71XmN?usp=sharing`

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Introduction

Speech Emotion Recognition (SER) is a fundamental problem in affective computing and human–computer interaction, where the objective is to infer emotional states from speech signals. Emotional information is conveyed through multiple acoustic characteristics, including pitch variation, vocal intensity, rhythm, spectral structure, and speaking rate, as well as through linguistic content.

Recent advances in deep learning have significantly improved SER performance through representation learning using convolutional, recurrent, and transformer-based architectures. However, datasets containing only a limited number of speakers remain highly vulnerable to speaker leakage under conventional random train–test splitting strategies. In such settings, models may partially exploit speaker-specific acoustic characteristics rather than learning speaker-invariant emotional representations, resulting in inflated evaluation performance.

This work investigates speech-only, text-only, and multimodal speech–text emotion recognition on the Toronto Emotional Speech Set (TESS) under speaker-aware evaluation settings. The study focuses on cross-speaker generalization and multimodal learning behaviour under weak semantic conditions, where emotional information is encoded predominantly through acoustic delivery rather than lexical content.

Three primary pipelines were evaluated:

- **Speech-only pipeline**: CNN-BiLSTM-Attention architectures and pretrained speech representations for acoustic emotion modelling,
- **Text-only pipeline**: DistilBERT-based contextual representations derived from filename-extracted transcript words,
- **Multimodal fusion pipeline**: projection-based fusion of acoustic and textual embeddings.

Experimental results demonstrate that acoustic representations dominate emotional separability in TESS, while the text-only pipeline remains near random-chance performance due to the absence of discriminative lexical semantics. The experiments further show that multimodal fusion remains predominantly speech-driven when the auxiliary textual modality contributes limited semantic information. The highest reported accuracies were achieved under limited target-speaker adaptation settings rather than strict fully unseen-speaker evaluation.

The primary contributions of this work are summarized as follows:

- rigorous speaker-aware evaluation of speech-only, text-only, and multimodal SER pipelines,
- analysis of speaker leakage and cross-speaker generalization under limited-speaker conditions,
- investigation of multimodal fusion behaviour under semantically weak textual settings,
- comparative evaluation of handcrafted, deep learning, and transformer-based speech representations.

1.1 Dataset

The experiments were conducted using the Toronto Emotional Speech Set (TESS), which contains 2,800 emotional speech recordings collected from two female speakers: OAF (Older Adult Female) and YAF (Young Adult Female). The dataset contains seven emotional categories: anger, disgust, fear, happiness, neutral, pleasant surprise, and sadness.

Each filename encodes the speaker identity, spoken word, and emotion label. For example, `OAF_back_angry.wav` represents the OAF speaker uttering the word "back" with the emotion label "angry". The filename-derived transcript words were used as the textual modality for the text-only and multimodal fusion experiments.

Architecture Decisions

This chapter presents the architectural design of the three primary emotion recognition pipelines evaluated in this study: speech-only, text-only, and multimodal fusion. The overall framework follows a sequential processing pipeline consisting of preprocessing, feature extraction, representation modelling, multimodal fusion, and final classification.

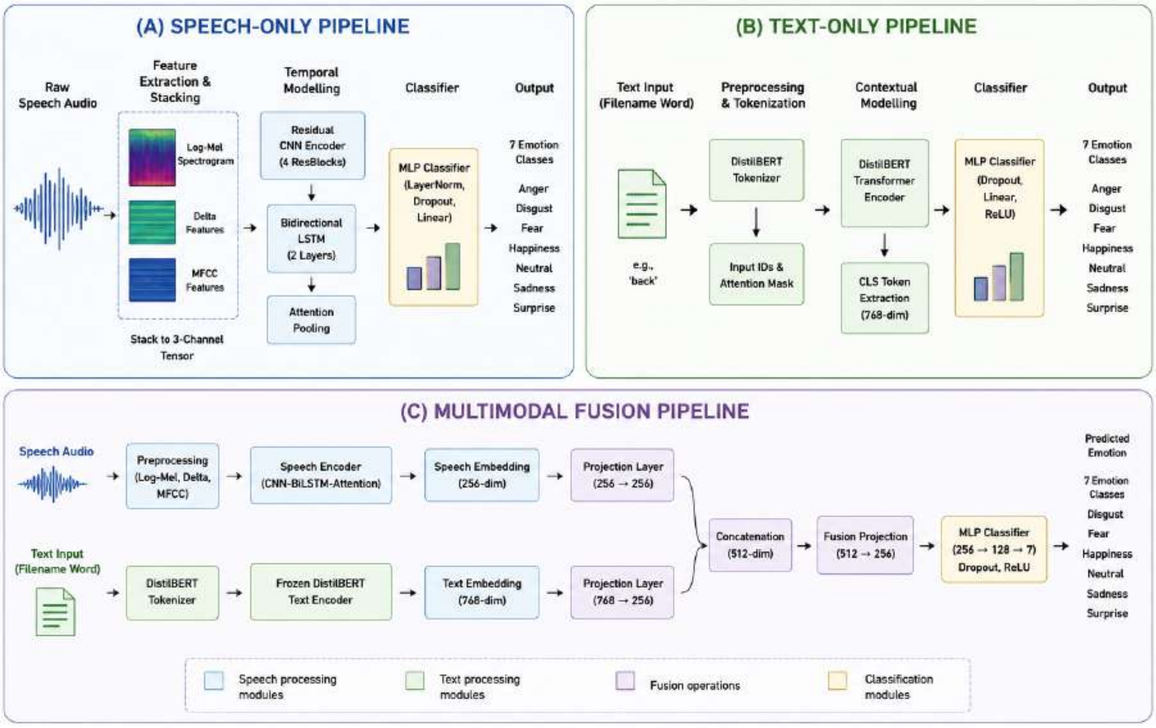

Figure 2.1: Overview of the speech-only, text-only, and multimodal fusion architectures evaluated in this study.

2.1 Speech-Only Pipeline

The speech-only pipeline was designed to learn emotion-discriminative acoustic representations directly from speech signals through hierarchical spectral feature learning, temporal sequence modelling, and attention-based aggregation.

2.1.1 Preprocessing

Audio recordings were standardized prior to feature extraction to reduce acoustic variability and improve optimization stability during training.

Table 2.1: Speech preprocessing configuration

Parameter	**Value**
Sampling rate	16 kHz
Audio duration	3 seconds
FFT window size	1024
Hop length	256
Mel bands	128
MFCC coefficients	40

A sampling rate of 16 kHz was selected to preserve speech-relevant frequency content while maintaining computational efficiency. Fixed-duration audio segmentation was used to ensure consistent tensor dimensions across batches.

2.1.2 Feature Extraction

Three complementary acoustic representations were extracted:

- Log-Mel spectrograms,
- delta features,
- MFCC representations.

The stacked multi-channel representation preserves spectral structure, temporal dynamics, and cepstral information simultaneously. These representations were combined into a unified acoustic tensor for downstream representation learning.

MFCC representations were dimensionally aligned with Mel-spectrogram features before multi-channel stacking.

2.1.3 Temporal Modelling

The acoustic tensor was processed using residual convolutional blocks with progressively increasing channel dimensions:

$$3 \rightarrow 32 \rightarrow 64 \rightarrow 128 \rightarrow 256$$

The convolutional encoder learns hierarchical spectral representations at multiple abstraction levels, while residual connections improve optimization stability in deeper architectures.The final convolutional feature maps were reshaped into sequential embeddings before Bidirectional LSTM processing, enabling temporal modelling across the extracted acoustic feature sequence.

The sequential embeddings generated by the CNN encoder were subsequently processed using a two-layer Bidirectional LSTM with hidden dimension 256 to model long-range temporal dependencies in emotional speech. An attention pooling layer was applied after recurrent modelling to emphasize emotionally informative temporal regions and suppress less discriminative segments.

2.1.4 Classifier

The final acoustic embedding was processed using fully connected layers with layer normalization and dropout regularization prior to softmax classification:

$$512 \rightarrow 256 \rightarrow 7$$

Dropout regularization was introduced to reduce overfitting under the limited-speaker evaluation setting.

2.2 Text-Only Pipeline

The text-only pipeline was implemented to evaluate the effectiveness of filename-derived transcript words for emotion recognition in TESS.

2.2.1 Preprocessing

The textual modality was constructed by automatically extracting carrier words directly from the TESS audio filenames (e.g., `OAF_back_angry.wav` → `back`). During preprocessing, the filename string was parsed to isolate the spoken carrier word while removing speaker identifiers and emotion labels. The extracted words were then normalized using lowercase conversion prior to tokenization.

The processed text was converted into token IDs and attention masks using the pretrained DistilBERT tokenizer with:

```
MAX_LEN = 32
```

Table 2.2: Text preprocessing configuration

Parameter	Value
Tokenizer	DistilBERT tokenizer
Maximum sequence length	32
Text source	Filename-derived carrier words
Text normalization	Lowercase conversion

A short maximum sequence length was sufficient because the textual modality consists only of isolated carrier words rather than full conversational utterances.

2.2.2 Contextual Modelling

The pretrained `distilbert-base-uncased` transformer encoder was used to generate contextual text representations. The CLS-token embedding extracted from the final transformer layer was used as the sequence-level representation for downstream classification.

DistilBERT was selected because it provides computationally efficient contextual embeddings while retaining most of the representational capacity of larger BERT-based architectures.

However, the TESS textual modality remains inherently weak because identical carrier words appear across multiple emotional categories. Consequently, the transformer encoder receives minimal emotion-discriminative semantic information during training, limiting the separability of the learned textual embedding space.

2.2.3 Classifier

The transformer embedding was processed using fully connected layers with ReLU activation and dropout regularization:

$$768 \rightarrow 256 \rightarrow 7$$

2.3 Multimodal Fusion Pipeline

The multimodal fusion pipeline was designed to investigate joint representation learning from acoustic and textual modalities under weak semantic conditions.

2.3.1 Feature Extraction and Contextual Modelling

The speech branch in the multimodal fusion pipeline uses a lightweight CNN-BiLSTM-Attention architecture optimized for multimodal training efficiency. The lightweight architecture was selected to reduce multimodal optimization complexity while preserving sufficient temporal modelling capacity for emotional speech representation learning.

The encoder consists of convolutional layers with channel progression $3 \rightarrow 32 \rightarrow 64 \rightarrow 128$, followed by a Bidirectional LSTM with hidden dimension 128 per direction.

The speech branch processes stacked Log-Mel, delta, and MFCC representations to learn hierarchical spectral and temporal emotional structure. The convolutional layers extract localized acoustic patterns, while the Bidirectional LSTM performs temporal modelling across sequential emotional dynamics and prosodic transitions. The attention layer further emphasizes emotionally informative temporal regions before multimodal fusion.

The text branch generates CLS-token embeddings from filename-derived carrier words using a pretrained DistilBERT encoder. DistilBERT was selected because it provides computationally efficient contextual embeddings while maintaining most of the representational capacity of larger transformer architectures.

During multimodal training, the DistilBERT encoder was frozen to reduce overfitting caused by repetitive and semantically weak textual inputs. Freezing the transformer branch additionally stabilized optimization by preventing noisy gradient updates from the weak textual modality.

2.3.2 Fusion

The speech and text embeddings were independently projected into a shared multimodal feature space using linear projection layers. The acoustic embeddings were mapped from $256 \rightarrow 256$, while the DistilBERT text embeddings were projected from $768 \rightarrow 256$ dimensions prior to fusion.

Projection layers were introduced to align the statistical distributions of acoustic and textual embeddings before multimodal fusion.

The projected embeddings were fused using feature concatenation followed by a fusion projection block consisting of linear projection, batch normalization, ReLU activation, and dropout regularization. Feature-level fusion was selected instead of decision-level fusion to enable joint multimodal representation learning prior to classification.

This design improves representation alignment while suppressing modality-specific noise introduced by the weak textual branch.

2.3.3 Classifier

The fused multimodal representation was processed using fully connected classification layers with dimensional progression $512 \rightarrow 256 \rightarrow 128 \rightarrow 7$ before final softmax prediction.

Progressive dimensionality reduction inside the classifier improves representation compression while preserving emotionally discriminative structure before final prediction.

The proposed framework was designed specifically to analyze modality informativeness, speaker-domain mismatch, and representation evolution under speaker-aware emotion recognition settings.

Training Configuration

All primary experiments were implemented using PyTorch and trained using the AdamW optimizer under speaker-aware evaluation settings. Early stopping and learning-rate scheduling were applied to improve optimization stability and reduce overfitting under limited-speaker conditions.

Table 3.1: Training configuration across the primary TESS pipelines

Configuration	**Speech Pipeline**	**Text Pipeline**	**Fusion Pipeline**
Architecture	CNN-BiLSTM-Attention	DistilBERT	CNN-BiLSTM-Attention + Frozen DistilBERT
Optimizer	AdamW	AdamW	AdamW
Learning Rate	1×10^{-4}	2×10^{-5}	2×10^{-5}
Batch Size	16	16	16
Loss Function	Cross-Entropy	Cross-Entropy	Cross-Entropy
Dropout	0.3	0.3	0.3
Scheduler	ReduceLR	None	None
Framework	PyTorch	PyTorch	PyTorch

The speech-only pipeline used a higher learning rate because the acoustic encoder was trained primarily from scratch using handcrafted feature representations. In contrast, the text-only and multimodal pipelines used smaller learning rates to preserve the stability of the pretrained transformer embeddings during optimization.

The multimodal fusion pipeline additionally froze the DistilBERT text encoder during training to reduce noisy gradient updates caused by the semantically weak textual modality. This stabilization strategy improved multimodal optimization consistency and reduced overfitting under limited-speaker evaluation settings.

Early stopping was monitored using validation Macro F1-score because class-sensitive evaluation remained more informative than overall accuracy under speaker-independent emotion recognition conditions.

Experiments

4.1 Evaluation Methodology

Evaluation methodology is particularly important in TESS-based emotion recognition because the dataset contains recordings from only two speakers. Conventional random train-test splitting introduces substantial speaker leakage, where samples from the same speaker appear in both subsets. Under these conditions, models may exploit speaker-specific acoustic cues instead of learning speaker-invariant emotional representations.

To obtain a more realistic evaluation of cross-speaker generalization, the experiments primarily used speaker-independent splitting. One speaker was used entirely for training, while the remaining unseen speaker was reserved for validation and testing. This setup provides a baseline measure of cross-speaker generalization.

Three evaluation strategies were investigated:

- **Speaker-Independent Evaluation:** Training and testing were performed on different speakers without target-speaker exposure during training.
- **Speaker Adaptation Evaluation:** A small percentage of target-speaker samples (5% or 15%) was introduced during training to reduce acoustic domain mismatch.
- **Random Speaker-Dependent Split:** Random train-test partitioning where both speakers appear across all subsets.

The random speaker-dependent setup produced near-perfect performance, but these results do not reflect true generalization because speaker information appears in both training and testing distributions. In contrast, speaker-independent evaluation caused performance degradation in several architectures, particularly for acoustically similar high-arousal emotions such as anger and happiness. Consequently, random speaker-dependent evaluation was used only as a reference baseline and not as a valid measure of cross-speaker generalization.

The adaptation experiments further showed that minimal target-speaker calibration data reduces cross-speaker mismatch and improves representation consistency. However, the resulting setup no longer represents strict zero-shot speaker-independent evaluation.

4.2 Speech-Only Experiments

The speech-only experiments were designed as a progressive investigation into acoustic representation quality under speaker-independent emotion recognition. The experimental sequence begins with handcrafted CNN-based acoustic modelling, followed by temporal sequence enhancement using CNN-BiLSTM-Attention architectures, and finally transitions toward pretrained transformer-based speech encoders and speaker adaptation strategies. Each stage was designed to address limitations observed in the previous configuration, enabling systematic analysis of representation robustness, speaker invariance, and cross-speaker generalization behaviour.

4.2.1 Baseline CNN-BiLSTM-Attention Pipeline

The baseline speech architecture used stacked Log-Mel spectrograms, delta features, and MFCC representations processed using residual CNN layers, Bidirectional LSTM temporal modelling, and attention pooling. This baseline configuration was designed to evaluate whether handcrafted multi-channel acoustic representations combined with temporal sequence modelling could learn speaker-invariant emotional structure under cross-speaker evaluation.

The CNN encoder learned hierarchical spectral representations associated with emotional prosody, while the Bidirectional LSTM captured long-range temporal dependencies in vocal delivery. Attention pooling further improved representation quality by emphasizing emotionally informative temporal regions.

Although the architecture achieved moderate cross-speaker generalization, substantial degradation remained under strict unseen-speaker evaluation. The learned representations continued to exhibit sensitivity to speaker-dependent pitch distributions and vocal characteristics, motivating investigation of more robust pretrained transformer-based speech representations in subsequent experiments.

4.2.2 CNN Architecture Variations

Multiple CNN-based architectural variants were evaluated to analyze how convolutional depth, attention mechanisms, and temporal sequence modelling influence speaker-independent emotional representation learning. These experiments were designed to identify whether performance limitations originated primarily from insufficient spectral feature extraction capacity, weak temporal modelling, or instability under cross-speaker acoustic variation.

Table 4.1: CNN-based architecture variations.

Architecture	Accuracy	Macro F1	Observation
Simplified CNN	62.43%	56.19%	Severe collapse on anger and sadness classes.
CNN + Attention	57.14%	50.46%	Training instability and weak happiness discrimination.
Unified Audio CNN	66.11%	62.82%	Improved feature learning with moderate generalization gains.
Deep CNN + Attention	63.86%	59.02%	Moderate cross-speaker performance.
Baseline Audio CNN	59.21%	52.11%	Weak speaker-independent robustness.
CNN-BiLSTM-Attention	64.71%	59.74%	Temporal modelling improved representation stability.
Strict Zero-Shot CNN-BiLSTM-Attention	61.43%	58.22%	Strong degradation under unseen-speaker evaluation.

The baseline CNN-BiLSTM-Attention experiments were evaluated under multiple speaker-independent configurations and random seed initializations. The stricter zero-shot configuration produced lower performance because no target-speaker calibration data was introduced during training.

Most CNN-based models struggled to generalize across speakers, particularly for high-arousal emotional categories. The learned representations frequently relied on speaker-dependent pitch distributions instead of stable emotional structure.

4.2.3 Transformer-Based Speech Models

Pretrained transformer-based speech encoders were evaluated to investigate whether large-scale self-supervised acoustic pretraining improves speaker-invariant emotional representation learning beyond handcrafted CNN-based pipelines. Unlike handcrafted acoustic features, transformer speech encoders learn contextual representations directly from large-scale speech corpora, enabling richer modelling of prosodic structure, temporal dynamics, and speaker-normalized acoustic patterns.

The pretrained wav2vec2-base model achieved 62.64% accuracy and 57.61% Macro F1 under zero-shot speaker-independent evaluation, demonstrating slightly stronger structural robustness than several handcrafted-feature CNN pipelines. After limited target-speaker adaptation, WavLM-base achieved 99.89% accuracy and Macro F1 under speaker-adapted evaluation settings.

The pretrained transformer encoders generated more robust acoustic embeddings than handcrafted feature pipelines under strict speaker-independent evaluation. However, substantial performance degradation remained under fully zero-shot conditions, indicating that improved representation quality alone is insufficient to completely eliminate speaker-domain mismatch. These findings motivated further investigation into limited speaker adaptation as a mechanism for reducing cross-speaker acoustic distribution shift.

4.2.4 Speaker Adaptation Experiments

Speaker adaptation experiments were conducted to investigate whether limited target-speaker exposure can reduce acoustic domain mismatch under speaker-independent evaluation. These experiments analyze the extent to which cross-speaker degradation originates from representation insufficiency versus speaker-distribution shift between training and testing conditions.

Table 4.2: Speaker adaptation experiments.

Model	Adaptation	Accuracy	Observation
CNN-BiLSTM-Attention	15%	99.52%	Resolved major cross-speaker confusion.
CNN-BiLSTM-Attention	5%	99.89%	Only one test sample misclassified.
WavLM-base	5%	99.89%	Substantial improvement after speaker adaptation using pretrained speech embeddings.

The adaptation experiments demonstrated that only minimal target-speaker calibration data was required to substantially improve emotional separability across speakers. Performance increased dramatically between the strict zero-shot and 5% adaptation settings, while further gains beyond 5% remained marginal. These findings suggest that the dominant limitation in TESS-based speaker-independent emotion recognition is not insufficient model capacity, but rather acoustic distribution mismatch between speakers.

4.3 Text-Only Experiments

The text-only experiments evaluated whether filename-derived transcript words contain sufficient semantic information for speaker-independent emotion recognition in TESS.

Table 4.3: Text-only experiments on TESS.

Model	Accuracy	Macro F1	Observation
DistilBERT	14.93%	7.29%	Performance remained near random chance.
BiGRU	13.86%	6.82%	Sequential modelling provided minimal improvement.

The textual modality in TESS consists only of isolated carrier words such as “back”, “bar”, and “goose”. Since identical words appear across multiple emotional categories, the transcripts contain minimal emotion-discriminative semantic information.

The DistilBERT pipeline collapsed primarily toward anger, disgust, and fear predictions while producing almost zero recall for happiness, neutral, sadness, and surprise. This behaviour indicates poor separability within the learned textual embedding space.

The BiGRU baseline produced similarly weak performance, indicating that the limitation originates primarily from the dataset rather than the architecture itself. Unlike conversational SER datasets, TESS encodes emotional information primarily through acoustic delivery rather than lexical semantics.

4.4 Multimodal Fusion Experiments

The multimodal fusion experiments combined acoustic and textual embeddings to analyze multimodal learning behaviour under unequal modality informativeness, where the speech modality remains substantially more discriminative than the textual modality.

Table 4.4: Multimodal fusion experiments on TESS.

Fusion Architecture	**Accuracy**	**Macro F1**	**Observation**
CNN-BiLSTM + DistilBERT + Concatenation	94.74%	94.67%	Unstable modality interaction.
CNN-BiLSTM + Frozen DistilBERT + Fusion Projection	98.60%	98.61%	Best custom fusion architecture.
WavLM + DistilBERT + Concatenation	99.68%	99.68%	Highest-performing pretrained fusion configuration.

The initial fusion architecture used direct concatenation between speech and text embeddings. Although the model achieved strong overall performance, the weak textual modality introduced additional feature-space variance during fusion.

To improve fusion stability, the DistilBERT encoder was frozen and projection layers were introduced before fusion. This improved optimization stability and produced more compact multimodal embeddings.

Despite the high fusion accuracy, the multimodal representation remained predominantly speech-driven because the textual modality contributed limited discriminative semantic information.

The fusion experiments should therefore be interpreted primarily as an analysis of multimodal learning behaviour under weak semantic conditions rather than as evidence of strong complementary multimodal representation learning.

4.5 Ablation Studies

Ablation experiments were conducted to analyze the contribution of acoustic feature stacking, speaker adaptation, and architectural design choices to overall system performance.

4.5.1 Feature Stacking Ablation

The effect of multi-channel acoustic feature stacking was evaluated by comparing single-channel Mel-spectrogram input against combined acoustic representations consisting of:

- Log-Mel Spectrograms
- Delta Features
- MFCC Features

The stacked representation improved speaker-independent performance by approximately 4%. The temporal derivative features provided additional transition dynamics and temporal motion information that improved emotional separability across acoustically similar classes.

4.5.2 Speaker Adaptation Ablation

Table 4.5: Ablation study of speaker adaptation ratios.

Adaptation	**Accuracy**	**Macro F1**	**Observation**
0%	61.43%	58.22%	Strong degradation under speaker mismatch.
5%	99.89%	99.89%	Major improvement after minimal adaptation.
15%	99.52%	99.52%	Only marginal gains beyond 5% adaptation.
Random Split	99.29%	99.29%	Performance likely influenced by speaker overlap between training and testing subsets.

4.5.3 Architecture Comparison

Table 4.6: Architecture comparison on TESS.

Architecture	**Adaptation**	**Accuracy**	**Macro F1**
CNN-BiLSTM-Attention	0%	61.43%	58.22%
wav2vec2-base	0%	62.64%	57.61%
CNN-BiLSTM-Attention	5%	99.89%	99.89%
WavLM-base	5%	99.89%	99.89%

The pretrained transformer encoders produced more robust baseline representations under zero-shot evaluation, while both pretrained and custom architectures achieved near-perfect performance after speaker adaptation. These results indicate that speaker mismatch, rather than architectural capacity, is the dominant limitation in TESS-based speaker-independent emotion recognition.

4.6 Comparative Analysis

Table 4.7: Final comparison across all TESS pipelines.

Pipeline	Accuracy	Macro F1	Observation
Speech-only	99.89%	99.89%	Highest performance under speaker-adapted evaluation.
Text-only	14.93%	7.29%	Near-random performance due to weak semantics.
Multimodal Fusion	98.60%	98.61%	Fusion remained predominantly speech-driven.

The experiments confirm that emotional information in TESS is encoded primarily through acoustic speech characteristics rather than lexical semantics. The results further indicate that multimodal fusion does not inherently guarantee improved emotional separability when one modality contributes limited discriminative information.

Although multimodal fusion achieved very high classification accuracy, the final multimodal representation remained dominated by the acoustic modality because the filename-derived textual input contributes minimal discriminative information.

Although the speaker-adapted pipelines achieved near-perfect classification accuracy, strict zero-shot speaker-independent evaluation produced substantially lower performance. These results indicate that speaker-domain mismatch remains one of the primary challenges in limited-speaker SER datasets such as TESS.

4.7 Final Findings

The experiments produced four primary findings:

1. Speaker-independent emotion recognition on TESS is highly sensitive to speaker-domain mismatch.
2. Even minimal target-speaker adaptation substantially improves cross-speaker representation stability.
3. Text-only emotion recognition fails on carrier-phrase datasets lacking meaningful semantic information.
4. Multimodal fusion remains effective only when auxiliary modalities contribute meaningful complementary emotional information.

Overall, the experiments confirm that TESS behaves fundamentally as an acoustic emotion recognition dataset, where emotional information is conveyed predominantly through vocal delivery rather than lexical content. The highest reported accuracies were obtained under speaker-adapted evaluation settings rather than strict zero-shot speaker-independent evaluation.

4.8 Limitations

Several limitations remain in the current study. First, TESS contains recordings from only two speakers, limiting demographic diversity and increasing sensitivity to speaker-domain mismatch. Second, the dataset consists of acted emotional speech recorded under controlled conditions rather than spontaneous conversational interaction. Third, the textual modality contains only isolated carrier words with minimal semantic variability, restricting the effectiveness of multimodal fusion. Consequently, the findings may not directly generalize to large-scale conversational SER datasets containing richer linguistic and contextual information.

Analysis

5.1 Which Emotions Are Easiest and Hardest to Classify?

The relative difficulty of the seven emotional categories was analyzed using the class-wise recall scores obtained from the final speech-only, text-only, and multimodal fusion pipelines.

Table 5.1: Class-wise recall comparison across the final pipelines.

Emotion	**Speech**	**Text**	**Fusion**
Anger	0.992	0.684	0.962
Disgust	1.000	0.323	0.970
Fear	1.000	0.038	0.977
Happiness	1.000	0.000	0.993
Neutral	1.000	0.000	1.000
Sadness	1.000	0.000	1.000
Surprise	1.000	0.000	1.000

The speech-only and multimodal fusion pipelines achieved highly stable classification performance across most emotional categories. Emotions such as Fear, Sadness, Neutral, and Surprise were the easiest to classify because they occupy relatively distinct acoustic regions inside the learned embedding space.

Fear produced highly separable embeddings because of its strong high-frequency spectral energy, irregular temporal dynamics, and elevated vocal tension. Similarly, sadness formed compact low-arousal clusters characterized by reduced vocal intensity, flatter intonation contours, and slower temporal variation. Neutral speech also remained highly separable because of its comparatively stable pitch distribution and low expressive variability.

Surprise and happiness achieved near-perfect recall but occasionally exhibited partial overlap because both emotions contain elevated pitch variation, expressive vocal energy, and strong high-arousal prosodic structure. Despite this similarity, the adapted speech representations remained sufficiently discriminative to preserve stable decision boundaries between the two classes.

The most difficult emotion to classify was anger. Although the final pipelines achieved highly stable overall performance, anger produced the only measurable recall degradation in both the speech-only and multimodal fusion systems. This behaviour originates from overlap between anger and other high-arousal emotional regions, particularly happiness and surprise. All three emotions contain elevated vocal intensity, expanded pitch range, and concentrated spectral energy distributions, increasing embedding-space similarity during speaker-independent evaluation.

The text-only DistilBERT pipeline behaved substantially differently from the acoustic pipelines. Multiple emotional categories including Happiness, Neutral, Sadness, and Surprise collapsed completely, producing zero recall. The learned semantic embedding space remained highly overlapped with almost no meaningful emotional separation.

This behaviour is expected because the TESS textual modality contains only isolated filename-derived carrier words repeated uniformly across emotional categories. Since the lexical content itself carries minimal emotion-discriminative semantics, the transformer encoder could not construct stable semantic decision boundaries regardless of modelling complexity.

The multimodal fusion pipeline achieved performance comparable to the speech-only architecture because the fused representation remained predominantly speech-driven. The weak textual branch introduced only limited feature-level variance and did not significantly alter the primary acoustic decision boundaries.

Overall, the experiments demonstrate that emotions with highly distinctive prosodic structure, spectral energy distribution, and temporal dynamics were the easiest to classify, whereas acoustically similar high-arousal emotions produced the strongest residual confusion. The analysis further confirms that TESS behaves fundamentally as an acoustic emotion recognition dataset where emotional separability is governed primarily by vocal delivery rather than lexical semantics.

5.2 When Does Fusion Help Most?

The effectiveness of multimodal fusion was analyzed by comparing the speech-only, text-only, and multimodal fusion pipelines under identical evaluation settings.

Table 5.2: Comparison of the final speech-only, text-only, and multimodal fusion pipelines.

Pipeline	**Accuracy**	**Macro F1**	**Primary Observation**
Speech-only	99.89%	99.89%	Strong acoustic separability across all emotional categories.
Text-only	14.93%	7.29%	Near-random semantic classification behaviour.
Multimodal Fusion	98.60%	98.61%	Stable multimodal learning with speech-dominated embeddings.

The multimodal fusion pipeline achieved stable classification performance, demonstrating that the CNN-BiLSTM-Attention speech encoder and the frozen DistilBERT text encoder were successfully integrated into a stable multimodal feature representation.

However, fusion did not significantly outperform the final speech-only pipeline. The speech-only architecture achieved marginally higher overall performance, indicating that the acoustic modality remained the dominant source of emotional information throughout the learning process.

This behaviour is expected because the TESS textual modality contains only short filename-derived carrier words with extremely weak emotional semantics. Since identical words appear across multiple emotional categories, the transformer embeddings contribute very limited discriminative information during fusion.

Despite the weak semantic modality, the fusion architecture remained highly stable because the DistilBERT encoder was frozen during training. Freezing the transformer branch prevented severe overfitting and reduced noisy semantic adaptation caused by repetitive text inputs.

The projection-based fusion strategy also improved multimodal representation quality by aligning the speech and text embeddings into a projected multimodal feature space before classification. Compared to direct concatenation, the projection layers produced more compact and stable decision boundaries across emotional categories.

Fusion was most effective for emotions already containing highly distinctive acoustic structure such as Fear, Disgust, Surprise, and Happiness. These emotions maintained highly stable recall after multimodal integration, indicating that the auxiliary text branch did not significantly disrupt the dominant acoustic embedding geometry.

Overall, the experiments demonstrate that multimodal fusion is most beneficial when:

- the dominant modality remains preserved,
- weaker modalities are carefully constrained,
- feature distributions are aligned before fusion,
- and the auxiliary modality does not introduce excessive representation noise.

The results further confirm that TESS behaves primarily as an acoustic emotion recognition dataset. The speech branch continued to define the dominant decision boundaries, while the textual modality acted mainly as a weak auxiliary feature source within the fused representation space.

Because the TESS textual modality contains only isolated carrier words, the multimodal fusion pipeline does not represent a fully semantically informative multimodal SER benchmark. Instead, the experiments primarily evaluate the stability of fusion architectures when one modality contributes limited discriminative information.

5.3 Detailed Error Case Analysis

To better understand the limitations of the learned acoustic and multimodal representations, several incorrect predictions from the speaker-independent and multimodal fusion experiments were analyzed. The selected failure cases demonstrate how overlapping prosodic structure, high-arousal similarity, and speaker-dependent acoustic variation influence the learned decision boundaries.

Table 5.3: Summary of major failure patterns observed across experiments.

True Emotion	Predicted Emotion	Primary Cause
Surprise	Happiness	Shared positive high-arousal acoustic structure and rising pitch trajectories.
Happiness	Surprise	Abrupt spectral transitions and expressive vocal energy overlap.
Sadness	Neutral	Low-energy acoustic overlap and flattened temporal contours.
Sadness	Anger	Speaker-dependent spectral bursts shifting embeddings toward high-arousal regions.

Most classification failures originated from overlap between acoustically similar emotional regions rather than complete representation collapse. High-arousal emotions such as Happiness, Surprise, and Anger produced the largest degree of confusion because all three emotions contain elevated vocal intensity, expanded pitch range, and strong spectral energy concentration.

5.3.1 High-Arousal Confusion

Case 1: `OAF_page_ps.wav`

True Label: Surprise **Predicted Label:** Happiness

The utterance contained rapid pitch elevation and strong positive vocal energy, producing acoustic structure closely resembling happiness. The learned embedding therefore shifted toward the happiness region because both emotions share highly expressive high-arousal prosody.

Case 2: `OAF_lease_happy.wav`

True Label: Happiness **Predicted Label:** Surprise

The sample exhibited abrupt spectral transitions, rising intonation, and concentrated upper-frequency energy. These properties generated an acoustic profile similar to surprise, causing partial overlap between the two expressive emotional categories.

Case 3: `OAF_wife_happy.wav`

True Label: Happiness **Predicted Label:** Surprise

Strong fricative energy near the final consonant boundary combined with elevated pitch variation produced sharp spectral transitions resembling surprise. This increased overlap with the neighbouring emotional region.

Case 4: `OAF_pool_ps.wav`

True Label: Surprise **Predicted Label:** Happiness

Unlike typical surprise samples containing abrupt vocal transitions, this utterance exhibited smoother pitch contours and more stable positive energy distribution. The resulting representation became more similar to happiness than surprise.

5.3.2 Low-Arousal and Speaker-Dependent Confusion

Case 5: `OAF_doll_sad.wav`

True Label: Sadness **Predicted Label:** Neutral

The utterance contained low vocal intensity, weak temporal variation, and flatter intonation contours. These characteristics strongly overlap with neutral speech, reducing separability inside the low-arousal embedding region.

Case 6: `YAF_fat_sad.wav`

True Label: Sadness **Predicted Label:** Anger

The younger speaker produced stronger consonant-level spectral bursts and increased vocal emphasis near the final phoneme boundary. This shifted the embedding toward the high-arousal region associated with anger, demonstrating the continued influence of speaker-dependent acoustic variation under speaker-independent evaluation.

Overall, the error analysis demonstrates that most failures were caused by overlapping prosodic structure and residual speaker-dependent acoustic variation rather than complete representation instability. High-arousal emotional categories produced the strongest embedding overlap, while low-arousal categories occasionally collapsed because of reduced spectral and temporal variation.

5.4 Representation Visualization and Embedding Analysis

To further analyze the learned feature representations, PCA-based visualization, confusion matrix analysis, and representation-evolution tracking were performed on the speech-only, text-only, and multimodal fusion pipelines. These experiments investigate how emotional separability evolves across different network stages and how modality informativeness influences the final embedding geometry.

The visualization experiments additionally provide qualitative interpretation of the learned decision boundaries under speaker-independent evaluation. Since TESS contains only two speakers recorded under controlled conditions, the resulting embedding distributions remain partially influenced by speaker consistency and limited acoustic variability.

5.4.1 System-Level Performance Comparison

The speech-only and multimodal fusion pipelines achieved highly stable overall performance, whereas the text-only pipeline remained close to random-chance behaviour because of weak lexical emotional information in TESS.

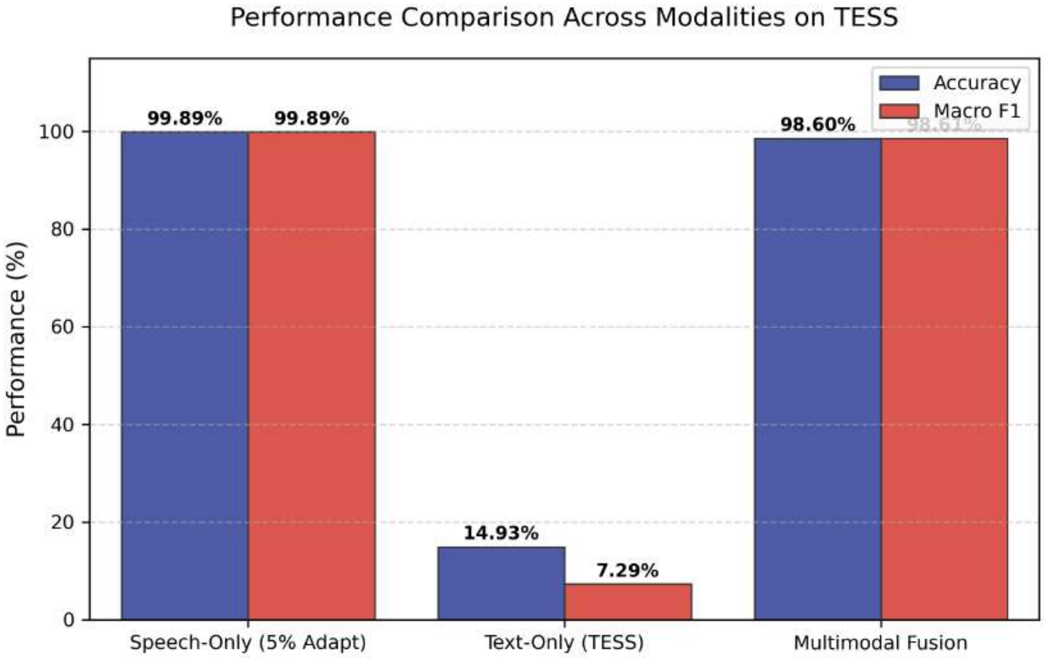

Figure 5.1: Performance comparison across speech-only, text-only, and multimodal fusion pipelines on the TESS dataset.

The comparison demonstrates that emotional separability in TESS is governed predominantly by acoustic information rather than lexical semantics. Although multimodal fusion achieved highly stable classification behaviour, the final performance remained slightly below the speech-only pipeline, indicating that the weak textual modality contributes limited complementary emotional information.

5.4.2 Confusion Matrix Analysis

Confusion matrix analysis was performed to identify residual overlap between emotional categories and to evaluate class-wise stability across the speech-only, text-only, and multimodal pipelines.

5.4.2.1 Speech-Only Pipeline

The speech-only pipeline produced strongly concentrated diagonal predictions with minimal residual inter-class overlap, indicating highly separable acoustic representations after speaker adaptation.

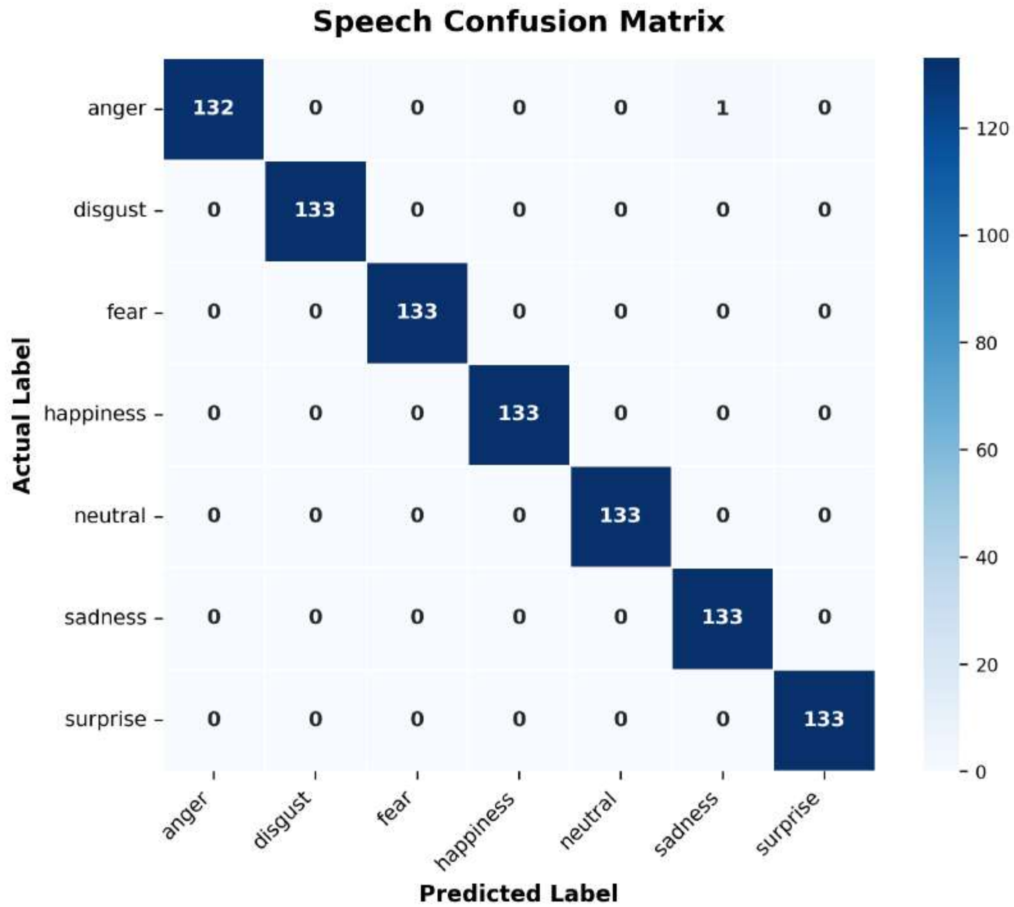

Figure 5.2: Confusion matrix of the final speech-only pipeline.

Most emotional categories achieved near-perfect classification consistency. Residual confusion occurred primarily between anger and sadness, where one anger sample shifted toward the sadness region. This behaviour reflects partial overlap between high-intensity vocal energy and lower-arousal temporal contours under speaker-independent evaluation.

The strong diagonal structure demonstrates that the CNN-BiLSTM-Attention architecture learned strongly structured acoustic embeddings after limited target-speaker adaptation.

5.4.2.2 Text-Only Pipeline

The text-only pipeline collapsed across multiple emotional categories and produced unstable semantic decision boundaries.

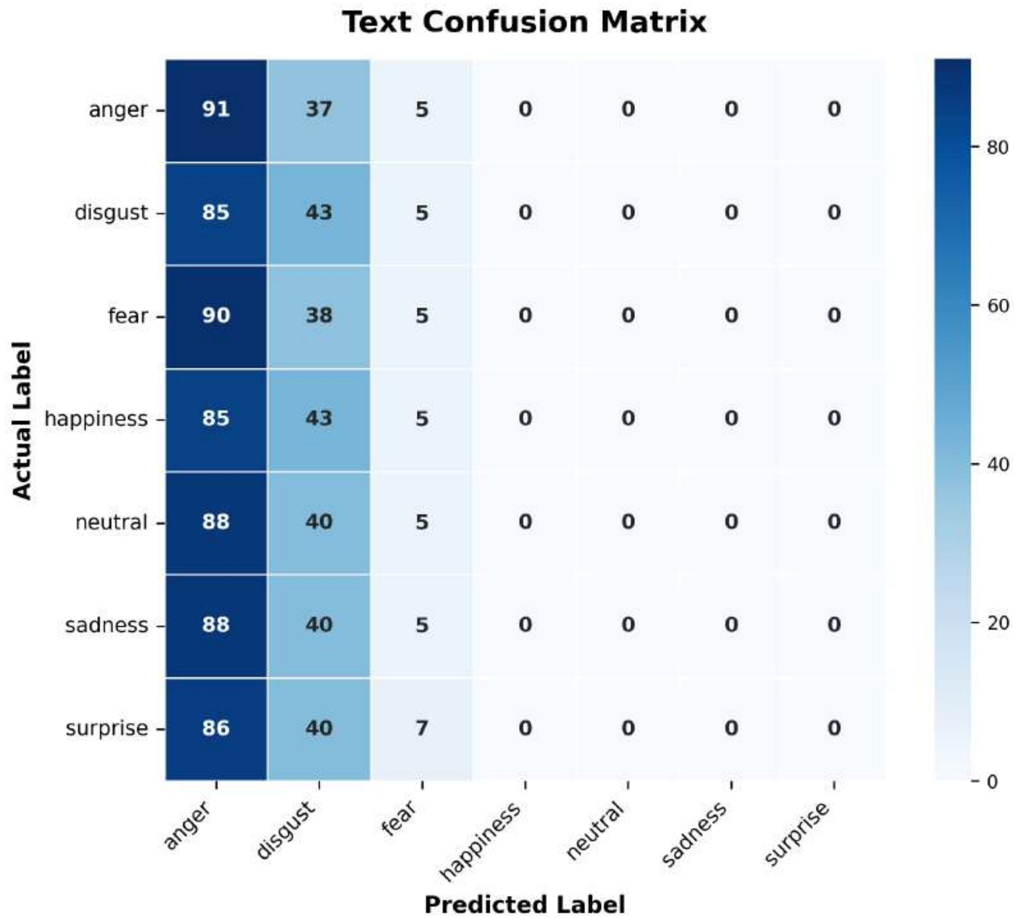

Figure 5.3: Confusion matrix of the DistilBERT text-only pipeline.

The model predicted most samples as anger or disgust while producing almost zero predictions for happiness, neutral, sadness, and surprise. This behaviour indicates severe embedding-space overlap and poor semantic separability.

The failure originates primarily from the dataset itself rather than the transformer architecture. Since identical carrier words appear uniformly across emotional categories, the textual modality contains almost no emotion-discriminative semantic structure. Consequently, the learned transformer embedding space remained highly collapsed throughout training.

5.4.2.3 Multimodal Fusion Pipeline

The multimodal fusion pipeline largely preserved the acoustic structure learned by the speech branch after multimodal integration.

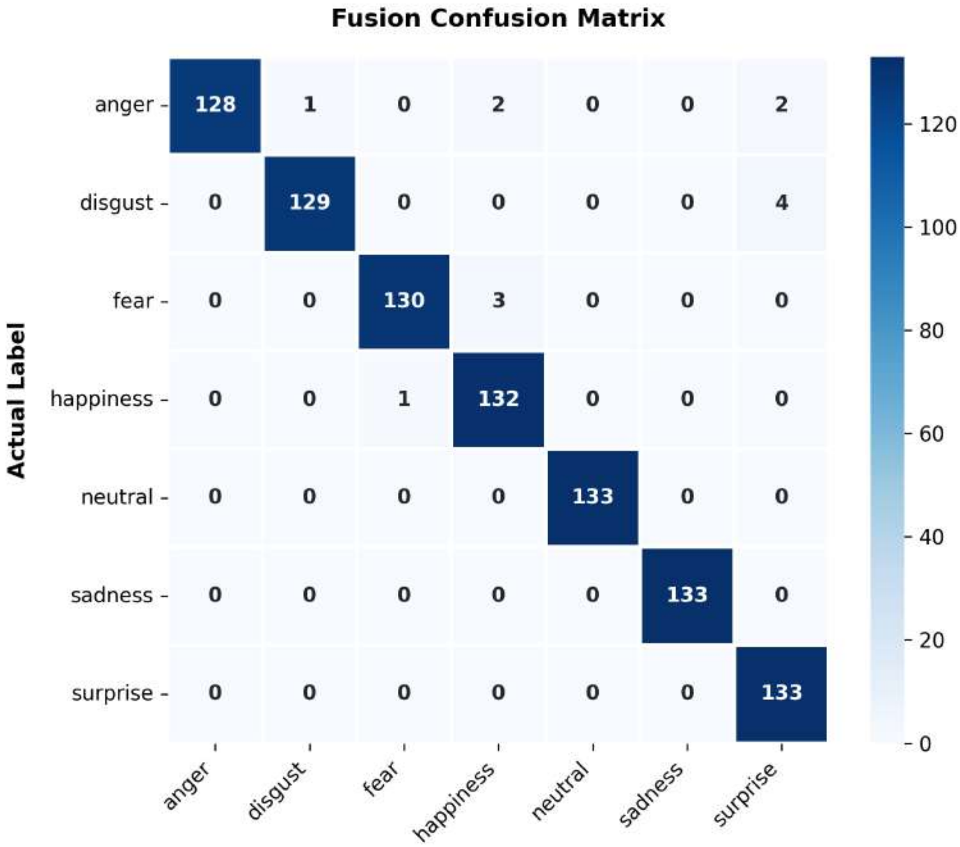

Figure 5.4: Confusion matrix of the multimodal fusion pipeline.

Most emotional categories remained highly stable after fusion. Residual confusion occurred primarily between anger, happiness, and surprise because these emotions occupy partially overlapping high-arousal acoustic regions.

Compared to the speech-only confusion matrix, the fusion model introduced slightly increased inter-class variance. However, the dominant acoustic decision boundaries remained preserved because the weak textual modality contributed only limited semantic influence during multimodal learning.

5.4.3 Embedding Space Visualization

PCA-based embedding visualization was performed to analyze emotional cluster geometry, inter-class overlap, and representation evolution across different modelling stages.

The compact cluster structures observed in several visualizations are influenced partly by the controlled recording environment and limited-speaker characteristics of the TESS dataset.

5.4.3.1 Speech Embedding Space

The speech embedding space forms highly compact and well-separated emotional clusters with minimal inter-class overlap.

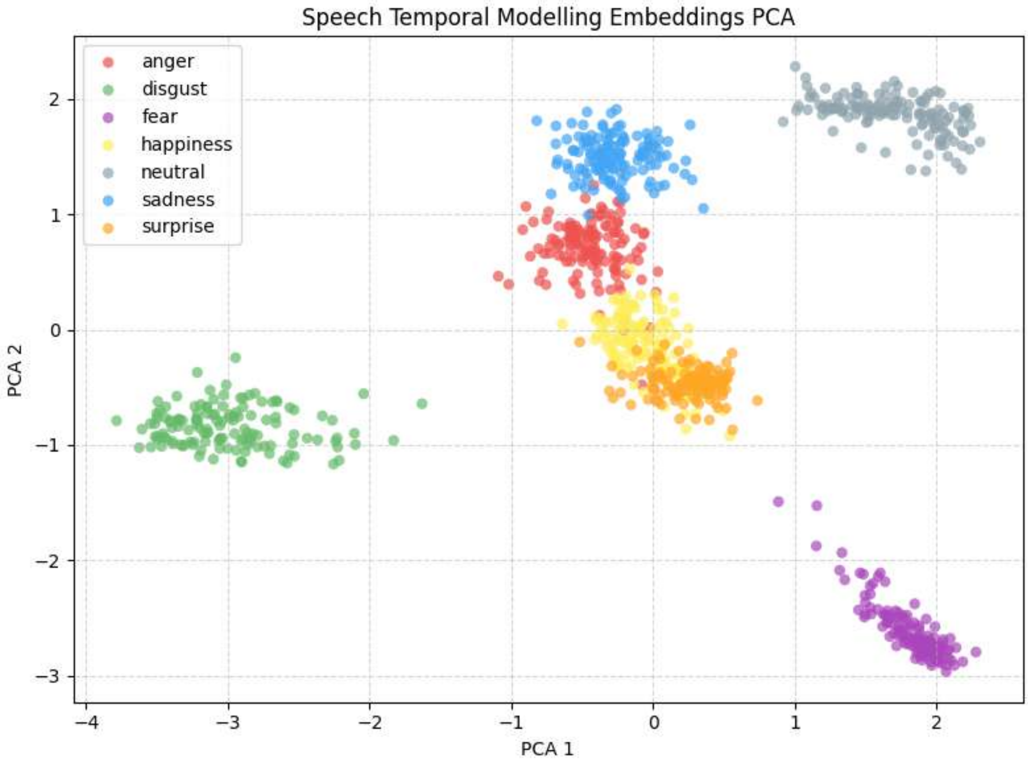

Figure 5.5: PCA visualization of the speech-only embedding space.

Distinct emotional regions emerged for disgust, fear, sadness, neutral, and surprise, demonstrating strong acoustic separability inside the learned representation space. Fear and sadness formed particularly compact clusters because of highly distinctive temporal dynamics and spectral structure.

The largest overlap occurred between happiness and surprise, where both emotions occupy neighbouring high-arousal acoustic regions characterized by elevated pitch variation and expressive vocal energy.

Overall, the visualization confirms that the speech encoder learned strongly structured acoustic manifolds capable of separating most emotional categories under speaker-independent evaluation.

5.4.3.2 Speech Representation Evolution

The speech-only architecture progressively improved emotional separability across successive network stages.

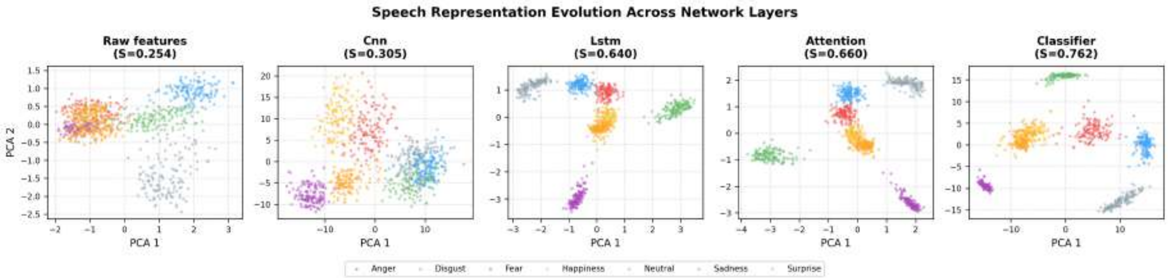

Figure 5.6: Evolution of speech embeddings across the CNN, BiLSTM, Attention, and classifier stages.

The raw acoustic features initially exhibited substantial inter-class overlap with weak emotional structure. After CNN processing, coarse emotional clustering began to emerge through hierarchical spectral feature extraction.

The BiLSTM stage significantly improved temporal organization by modelling long-range emotional prosody and sequential vocal dynamics. The attention layer further compressed emotionally informative regions into more compact and separable clusters.

The final classifier embeddings produced the strongest emotional separation, indicating that the network progressively transformed low-level spectral information into highly discriminative emotional representations.

5.4.3.3 Text Embedding Space

The text embedding space remained highly collapsed with severe inter-class overlap.

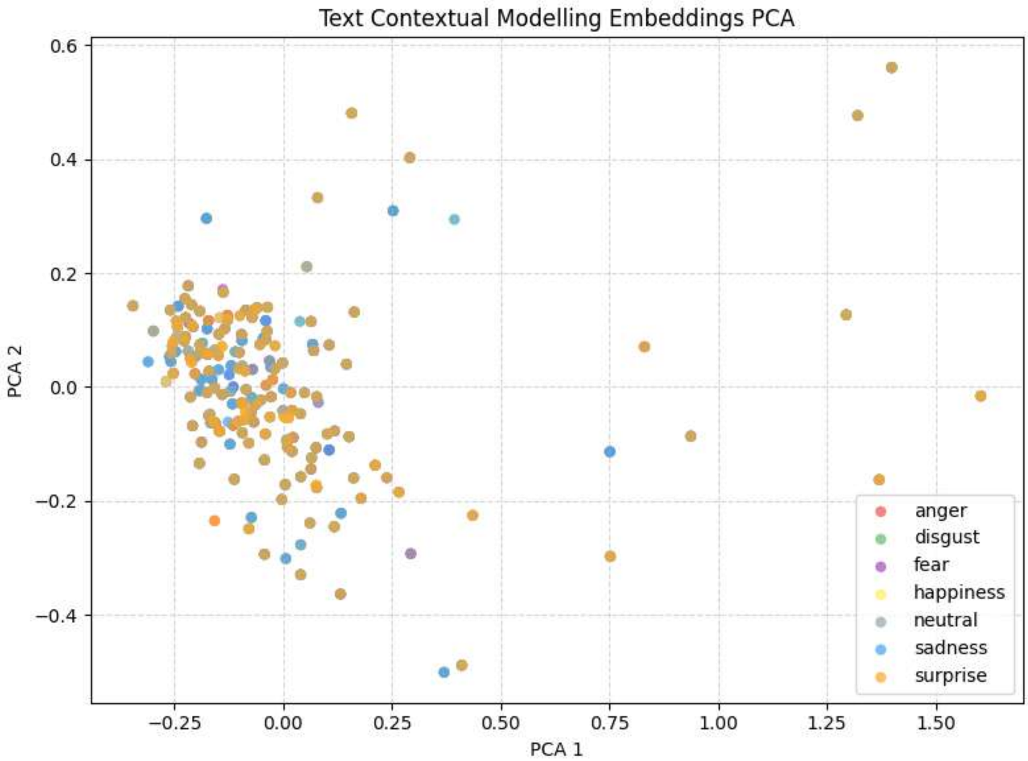

Figure 5.7: PCA visualization of the text-only embedding space.

Unlike the speech embeddings, the textual representations failed to form meaningful emotional clusters. Most samples occupied a single highly overlapped semantic region with minimal class-wise separation.

This behaviour confirms that the isolated TESS carrier words contain insufficient semantic variability for reliable emotion recognition. Since the lexical content remains nearly identical across emotional categories, the transformer encoder could not construct stable emotion-discriminative semantic boundaries.

5.4.3.4 Text Representation Evolution

The text representation evolution remained highly unstable across all network stages.

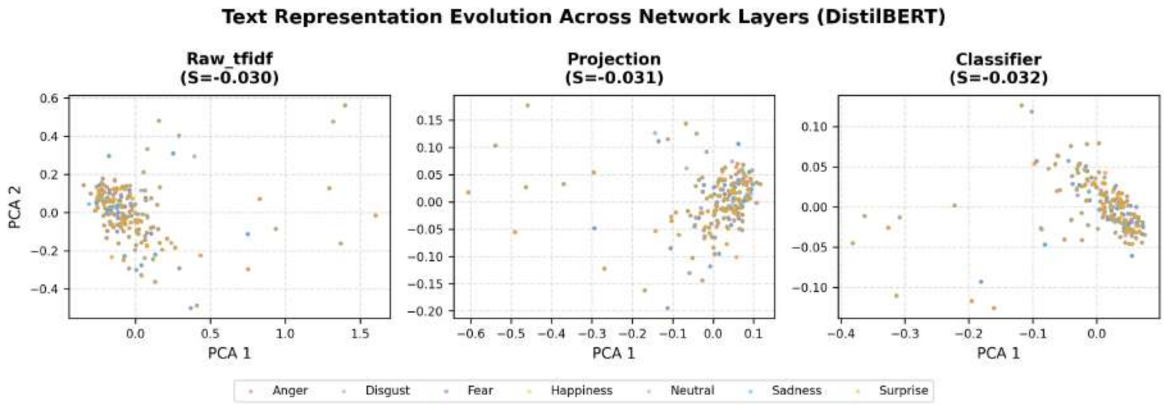

Figure 5.8: Evolution of text embeddings across the raw TF-IDF, projection, and classifier stages.

The raw textual features initially exhibited highly overlapped semantic distributions. Even after projection and classifier transformation, the embedding geometry remained weakly separable with no clearly defined emotional manifolds.

Unlike the speech pipeline, increasing representational depth did not substantially improve class separability because the underlying modality itself lacks meaningful emotion-discriminative information.

5.4.3.5 Fusion Embedding Space

The multimodal fusion embedding space largely preserved the dominant acoustic structure learned by the speech branch while integrating limited auxiliary textual information.

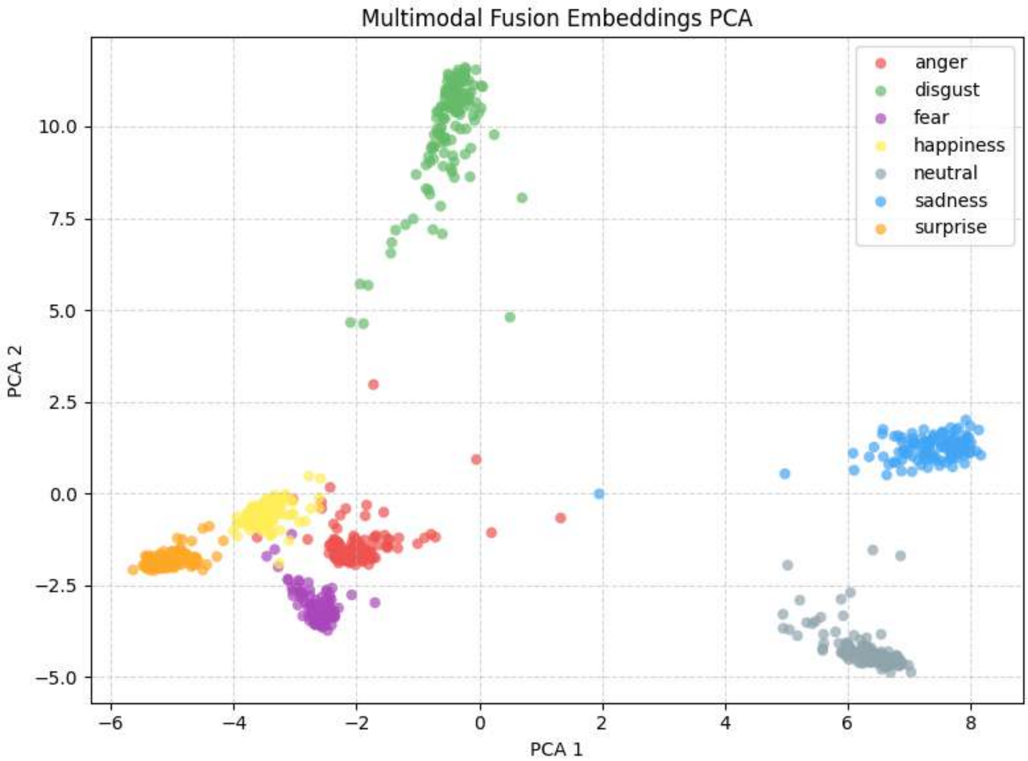

Figure 5.9: PCA visualization of the multimodal fusion embedding space.

Most emotional categories remained compact and highly separable after multimodal integration. The resulting cluster geometry strongly resembles the speech-only embedding distribution, confirming that the final fused representation remained predominantly speech-driven.

Minor increases in inter-class variance appeared near the anger and happiness regions after fusion, indicating that the weak textual modality introduced limited additional feature-space noise without substantially improving emotional separability.

5.4.3.6 Fusion Representation Evolution

The multimodal representation evolution demonstrates how the speech and text branches contribute differently to the final fused embedding geometry.

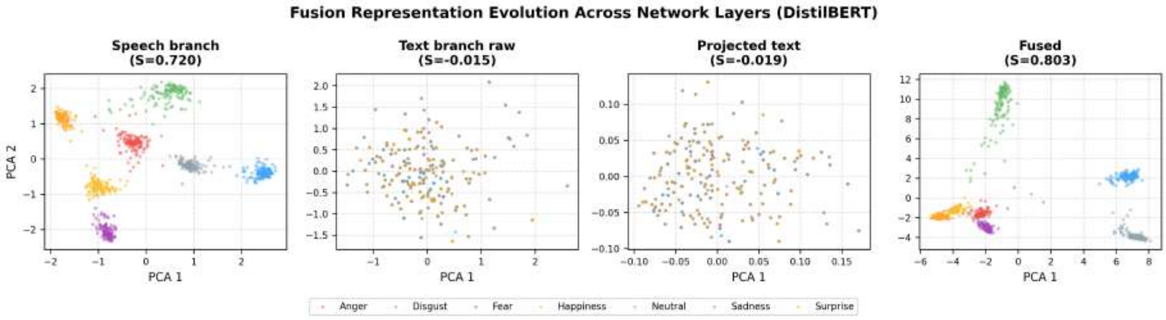

Figure 5.10: Evolution of multimodal fusion embeddings across the speech branch, text branch, projection layers, and fusion block.

The speech branch generated highly structured and separable emotional embeddings, whereas the text branch remained substantially overlapped because of weak semantic variability.

The projected text representations remained highly compressed with minimal emotional structure even after multimodal projection alignment. After fusion, the final embedding space largely inherited the structured geometry produced by the acoustic branch.

These findings confirm that the fusion architecture remained acoustically dominated throughout training. The textual modality contributed only limited auxiliary information and did not substantially alter the primary emotional decision boundaries learned from speech.

5.4.4 Training Curve Analysis

Optimization behaviour across the three pipelines was further analyzed using training-loss progression and validation performance curves.

The speech-only and multimodal pipelines converged stably with strong validation performance, whereas the text-only pipeline showed limited optimization progress because of weak semantic separability.

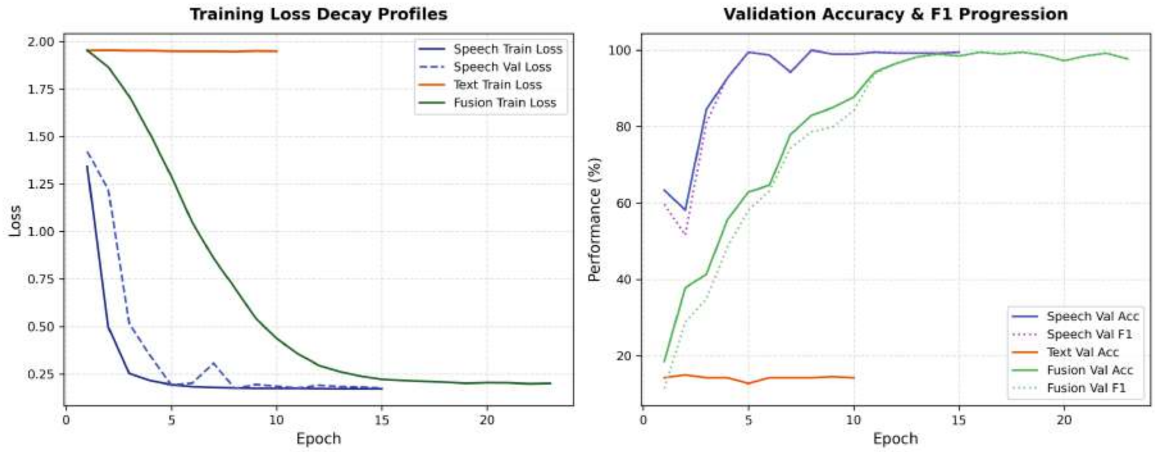

Figure 5.11: Training loss decay and validation performance progression across the speech-only, text-only, and multimodal fusion pipelines.

The speech-only pipeline demonstrated rapid convergence with highly stable validation Macro F1 after only a few epochs, indicating strong acoustic separability after speaker adaptation.

The multimodal fusion pipeline converged more gradually because of multimodal optimization complexity and projection alignment between acoustic and textual embeddings. Nevertheless, the final validation performance remained highly stable after convergence.

In contrast, the text-only pipeline exhibited almost no meaningful optimization improvement throughout training. The validation Macro F1 remained near random-chance behaviour despite stable loss progression, confirming that the semantic modality itself lacks sufficient discriminative information for reliable emotion classification.

Overall, the visualization and representation-analysis experiments demonstrate that emotional separability in TESS is governed primarily by acoustic structure, speaker-domain variability, and modality informativeness rather than architectural complexity alone.

5.4.5 Supporting Validation Experiments

To further evaluate whether the observed behaviour was specific to the TESS dataset, additional supporting experiments were conducted using the DailyDialog and MELD datasets. These experiments were designed to analyze how richer semantic content, conversational structure, speaker diversity, and real-world acoustic variability influence emotion recognition performance across text-only and multimodal settings.

Table 5.4: Supporting validation experiments on external conversational datasets.

Dataset	Pipeline	Accuracy	Macro F1	Observation
DailyDialog	RoBERTa-base Text-Only	77.34%	46.49%	Substantially improved semantic separability due to meaningful conversational text.
MELD	Speech-Text Fusion	59.50%	41.36%	Performance reduced because of multi-speaker variability, spontaneous dialogue, background noise, and class imbalance.

Unlike TESS, the DailyDialog dataset contains semantically meaningful conversational utterances with substantially richer emotional context. Consequently, the RoBERTa-base text pipeline achieved dramatically stronger performance compared to the near-random text-only behaviour observed on TESS. This result confirms that the poor TESS text performance originates primarily from dataset limitations rather than transformer modelling failure.

The MELD multimodal experiments produced substantially lower performance than the controlled TESS experiments because MELD contains realistic conversational audio with multiple speakers, spontaneous speech patterns, overlapping dialogue, background noise, and severe class imbalance. Emotional boundaries in MELD therefore become substantially less separable than in controlled acted datasets such as TESS.

These supporting experiments reinforce the central findings of the study regarding modality informativeness, speaker variability, and dataset-dependent emotional separability. The results further demonstrate that multimodal emotion recognition performance depends strongly on dataset characteristics, semantic richness, acoustic variability, and evaluation protocol design.

Conclusion

This work investigated speech-only, text-only, and multimodal speech–text emotion recognition on the TESS dataset under speaker-aware evaluation settings. The experiments demonstrated that emotional information in TESS is encoded primarily through acoustic speech characteristics rather than lexical semantics. The speech-only pipelines achieved the strongest performance, while the text-only models remained near random-chance level because the filename-derived carrier words contain minimal emotion-discriminative semantic information. The multimodal fusion experiments further showed that stable multimodal learning is achievable under weak semantic conditions, although the fused representations remained predominantly speech-driven. The study also confirmed that speaker-domain mismatch is one of the primary challenges in limited-speaker SER datasets, while even minimal target-speaker adaptation substantially improves cross-speaker representation stability. Overall, the findings highlight the importance of evaluation protocol design, modality informativeness, and speaker adaptation in emotion recognition systems.